Joseph Gruber

813.756.4705 (m) ✦ joseph.gruber@gmail.com ✦ U.S. Citizen

SPACECRAFT/SATELLITE ENGINEERING LEADER, SYSTEMS ENGINEER, SOFTWARE ENGINEER

Highlights

- Technical and programmatic leader in the aerospace industry leading multi-disciplinary teams across the system development lifecycle for complex, large-scale mission-critical engineering projects including satellite constellation mission planning software for NASA and international partners earth science constellation, flight dynamics ground control system for Aqua/Aura/Terra spacecraft, and flight software for Blue Origin's New Shepard human spaceflight vehicle.
- Motivated by a passion for combining the latest innovations, modern technology, and proven software development practices to solve the engineering challenges of the space industry, I thrive on leading passionate teams to deliver unique capabilities, a bias for action, and the use for automation and continuous learning to drive "agile aerospace".
- Widely praised for leading NASA Earth Science Mission Operations (ESMO) through an agile transformation on a flight operations ground control system, championing and implementing continuous integration to satellite engineering operations and product development, resulting in improved stakeholder communication, on-time deliveries, and enhanced system performance.
- Designed and implemented a comprehensive system integration and test plan to conduct verification, validation, and readiness activities to support flight mission and spacecraft operations while communicating to technical and non-technical stakeholders (NASA mission management team, mission users, contract staff, etc.)
- Strong proficiency and experience in planning and documenting requirements, driving product schedules, identifying and managing risk, communicating status across all levels of stakeholders, and balancing resource demands across multiple projects on highly integrated engineering teams.
- Specialized expertise in the areas of: spacecraft, low earth orbit (LEO) and geostationary (GEO) satellites, constellation management, ground control systems, systems integration, cloud computing and infrastructure, technical program management, systems and software engineering, CMMI-DEV Level 3, AS9100, team leadership, collaboration, innovative engineering solutions, system design and development, risk management, process improvement, change management, and growing & mentoring staff.

Professional Experience

Blue Origin, Technical Project Manager (2019 – present)

***New Shepard**: Human spaceflight software architecture, systems development, technical program management*

- Providing technical and programmatic engineering leadership and direction for 30+ software, network, and integration engineers across six multi-disciplinary engineering teams (GNC, Software Integration, Avionics Software, Hardware in the Loop lab, Verification, and Flight Network).
- Directed priority and guided definition, development, testing, and deployment of the system architecture, flight software, and ground network for the New Shepard human spaceflight system.
- Collaborated with internal and external stakeholders to define and decompose requirements, inter-dependencies, and system & software deliverables. Drove coordination with operations & maintenance team for continuous testing and deployment of flight software and integrated ground system software and procedures.
- Documented and instituted workflows to manage scope for multiple missions and coordinated multiple software release schedules to ensure required scope was delivered for the planned flight manifest.
- Created release roadmaps aligned to business objectives, tracked variances to schedule, and drove recovery plans to meet flight dates and objectives.
- Influenced subsystems to transition from waterfall methodology to Scrum resulting in improved release predictability, greater transparency across stakeholders, and ability to collect metrics and measure productivity across the products.
- Refined and communicated key program metrics including engineering schedules, procurement data, budget, resource supply/demand profiles, and program management status reports to senior leadership. Developed custom built, automated data reporting tools and dashboards to report key program performance data.

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a.i. solutions (NASA), Consulting Systems Engineer/Task Lead; Project Engineer/Functional Lead (2016 – 2019)

Space Network Ground Segment Sustainment (SGSS) *System engineering research, sustainment planning, modernization*

- Drove system level architecture design and advised on system/software engineering best practices for the modernization and sustainment of the Tracking and Data Relay Satellite System (TDRSS) ground terminals and space network operations center.
- Reviewed contractor system design and architecture documentation to identify lifecycle supportability risks and associated mitigation strategies and assess post-final acceptance review/handover operational posture.
- Performed engineering analyses and trade studies, formulating and documenting findings resulting from sustainment research of legacy systems and software. Created a configuration control system for long-term planning and asset reporting.

Earth Science Mission Operations *System/software engineering management, technical innovation, spacecraft operations*

- Responsible for engineering management, technical leadership, and program management for a two 15+ multi-disciplinary system & software engineering team supporting telemetry, tracking, and control (TT&C) of multiple spacecraft and missions including NASA's Aqua, Aura, and Terra. Developed and executed project management plan incorporating schedule, resources, risk, cost, quality, and configuration management.
- Designed, planned, and implemented automated debris avoidance maneuver capability, automated acquisition of spacecraft ephemeris data from the Joint Space Operations Center (JSpOC), and automated close approach violation notifications, increasing spacecraft safety awareness and improving efficiency across the flight team and external mission partners.
- Actively identified opportunities for technical and business process improvements instituting internal standards reflecting engineering best practices through standardized workflows, mandatory peer reviews, and documented standard operating procedures enabling automated testing and ensuring long-term maintainability of flight operations software.
- Interfaced with internal and external mission management and flight operation teams to maintain an environment of technical excellence and customer service ensuring successful mission support. Established technical interchange increasing communication between the engineering team and mission teams from multiple space agencies (JAXA, CNES, ESA, NASA).
- Integrated flight dynamics analysis team into system development lifecycle allowing increased collaboration and innovation with the system engineering and software engineering teams. Fostered a positive team culture of curiosity, collaboration, and mutually challenging team communication. Developed, mentored, and coached the engineering team to enhance productivity, increase morale, and ensure team success.

NASA / University of Hawaiʻi at Mānoa, Mission Support Lead (2013 – present)

Hawaii Space Exploration Analog & Simulation *Human spaceflight research, proto-flight system & infrastructure development*

- Serve as HI-SEAS mission support lead guiding 30+ experienced mission support team members on long-duration, isolated human spaceflight simulations ranging in duration from four months to one year.
- Direct the primary communication interface to crew members participating in analog simulations, coordinating and monitoring Mars simulated time-delayed mission operations, troubleshooting issues, and providing day-to-day operational support and technical expertise for distributed infrastructure (systems, network, sensor, power, water) at the habitat.
- Design and develop sensor monitoring software, backend services, and applications to interface between the HI-SEAS habitat and mission support systems. Produced recommendations on the usage of Amazon Web Services (AWS) EC2, S3, SES, & Lambda as components of mission management and Mars simulated time-delayed communication software infrastructure.

ECG, Inc. (NASA), Lead Planner (2014 – 2016)

Geostationary Operational Environmental Satellite-R Series (GOES-R) *IV&V, project management, spacecraft ground systems*

- Guided cross-functional planning coordination across multiple contractors, federal agencies, flight team, ground team, and post-handover team during deployment, integration, and verification & validation of disaggregate ground control systems, including software systems, ground segment (antenna system, RF hardware, etc.), and product distribution storage services, for the GOES-R geostationary weather satellite fleet. Built, maintained, and updated program/project integrated master plan (IMP) and integrated master schedule (IMS).
- Developed custom metrics analysis toolset, including performance trending and variance analysis, enabling current and historical assessment of baseline performance and actual execution. Contributed systems engineering expertise to the design and implementation of level zero storage system for petabyte-scale long-term baseline product data storage.
- Generated timelines for Launch and Orbit Raising (LOR) activities to verify on-orbit performance requirements and identified post-launch test events to validate the operational readiness of mission products. Integrated with supplemental near earth network (NEN) ground stations to establish requirements for LOR.

- Identified gaps in existing operational processes and procedures during verification and validation test events. Documented best practices and standard operating procedures ensuring compliance with flight project directives, NASA and NOAA policies, and federal government regulations.
- Supported chief engineer's office on flight readiness activities including performing and organizing mission readiness review, flight readiness review, and launch readiness review artifacts. Created program and project critical path analysis and contractor performance analyses in support of flight readiness.

Wyle (Naval Air Systems Command/NAVAIR), Lead Planner (2012 – 2014)

F-35 Joint Strike Fighter *Autonomous systems, Monte Carlo risk simulation, system development, aircraft sustainment*

- Mentored and led four-person planning team, along with integrated project teams, through project plan development, technical interchange meetings, integrated baseline reviews and program management reviews during system design and development (SDD) flight testing and initial operational test and evaluation (OT&E).
- Witnessed and supported test event dry runs, hardware-in-the-loop simulations, and for the record validation exercises reporting results and metrics to senior project leadership and adjusting future test events as required.
- Interfaced with project engineers, program managers, and analysts throughout all phases of the system development life cycle for the F-35 Autonomous Logistics Information System offboard mission support system.
- Conducted probabilistic risk assessments to model and determine the probability of completion of event milestones utilizing statistical analysis, including Monte Carlo simulations, to quantify technical and programmatic risks. Developed innovative, automated technical solutions to allow for improved data analytics of multiple, extremely large data sets.

CSC (Department of Defense), Technical Project Manager (2011 – 2012)

Engineering management, configuration management, requirements gathering and analysis, cross-agency collaboration

- Managed a team of five system engineering personnel, leading product development for a joint-agency communications and data transfer application and system. Responsible for the full engineering lifecycle including software and system requirements definition, design, development, test, and transition to operations and maintenance. Reduced six-month baseline slip to an on-time delivery for twelve remote site installations.
- Orchestrated technical leadership between the project management office (PMO) and regional technical staff contributing as subject matter expert and ensuring communication of stakeholder priorities across multiple teams within a matrixed, multiple contractor organization.

Prior Experience

- Self Employed (US Army, DoD – Connected Logistics, CompuGain Staffing), Technical Project Manager, 2010 – 2011
- SAIC / CTSC (Naval Station / Joint Task Force Guantanamo Bay), System Integration Engineer, 2009 – 2010
- Lockheed Martin, System Engineer, 2005 – 2008
- Rissman, Weisberg, Barrett, Hurt, Donahue, & McLain, P.A., Senior System Administrator, 2003 – 2005
- America Online, Senior Technical Consultant, 2001 – 2003

Education

- Embry-Riddle Aeronautical University; Masters in Aeronautical Science (Space Studies, Aerospace Management)
- Florida Institute of Technology; Bachelors in Business Administration, Computer Information Systems

Certifications

- Project Management Professional (PMP) – Project Management Institute (PMI)
- Professional Scrum Master I (PSM-I) – Scrum.org
- Satellite Tool Kit (STK) Master Certified – Analytical Graphics, Inc. (AGI)
- AWS Certified Cloud Practitioner

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Awards

- Received NASA Flight Dynamics Support Services Team Technical Innovation Award for engineering process improvements, including implementation of test-driven development, continuous integration and automated testing resulting in an 87% savings in manual effort by the flight operations team.
- Received NASA/NOAA GOES-R Process Improvement and Innovation Award for implementation of mission planning software solution allowing for improved performance monitoring across ground and flight project segments leading up to GOES-S Flight Readiness Review and GOES-R Launch Readiness Review.

Professional Affiliations

- Consultative Committee for Space Data Systems (CCSDS) – Associate
- PMI; PMBOK 5th Edition – Risk Management Content Editor
- American Institute of Aeronautics and Astronautics (AIAA) – SciTech Conference Chair
- AIAA National Capital Section (NCS) – Communications Committee
- International Council on Systems Engineering (INCOSE)
- Toastmasters – President, Advanced Communicator, Advanced Leader
- Civil Air Patrol, United States Air Force Auxiliary – Captain; Communications Officer (VHF/HF)

Technical Proficiencies

Systems: Windows Server, Linux (Ubuntu, Red Hat), RTLinux, Mac OS X, Solaris, VMware ESXi / vSphere, core Flight System (cFS), Amazon Web Services (Compute, Database, Networking, Storage)

Software: Atlassian (JIRA, Confluence, Bitbucket), IBM Rational DOORS, IBM Rational Team Concert (RTC), Oracle Primavera P6, Oracle Primavera Risk Analysis (PRA), Microsoft Project, Microsoft SQL Server, MySQL, MongoDB, MathWorks MATLAB, MathWorks Simulink, a.i. solutions FreeFlyer, AGI STK, Gitlab, GitHub, Jenkins, TeamCity

Programming Languages: Python, JavaScript, Node.js, PHP, C#, Visual Basic .NET, Visual Basic for Applications (VBA), Bash, C++